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Root filesystem organization

- ▶ The organization of a Linux root filesystem in terms of directories is well-defined by the **Filesystem Hierarchy Standard**
- ▶ <https://refspecs.linuxfoundation.org/fhs.shtml>
- ▶ Most Linux systems conform to this specification
 - Applications expect this organization
 - It makes it easier for developers and users as the filesystem organization is similar in all systems



Important directories (1)

/bin

Basic programs

)

/boot

Kernel images, configurations and initramfs (only when the kernel is loaded from a filesystem, not common on non-x86 architectures)

)

/dev

Device files (covered later)

)

/etc

System-wide configuration

)

/home

Directory for the users home directories

)

/lib



Important directories (1)

Basic libraries

)

/media

Mount points for removable media

)

/mnt

Mount point for a temporarily mounted filesystem

)

/proc

Mount point for the proc virtual filesystem

)



Important directories (2)

/root

Home directory of the **root** user

)

/run

Run-time variable data (previously **/var/run**)

)

/sbin

Basic system programs

)

/sys

Mount point of the sysfs virtual filesystem

)

/tmp

Temporary files

)

/usr

/usr/bin



Important directories (2)

Non-basic programs

)

/usr/lib

Non-basic libraries

)

/usr/sbin

Non-basic system programs

)

)

/var

Variable data files, for system services. This includes spool directories and files, administrative and logging data, and transient and temporary files

)



Separation of programs and libraries

- ▶ Basic programs are installed in `/bin` and `/sbin` and basic libraries in `/lib`
- ▶ All other programs are installed in `/usr/bin` and `/usr/sbin` and all other libraries in `/usr/lib`
- ▶ In the past, on UNIX systems, `/usr` was very often mounted over the network, through NFS
- ▶ In order to allow the system to boot when the network was down, some binaries and libraries are stored in `/bin`, `/sbin` and `/lib`
- ▶ `/bin` and `/sbin` contain programs like `ls`, `ip`, `cp`, `bash`, etc.
- ▶ `/lib` contains the C library and sometimes a few other basic libraries
- ▶ All other programs and libraries are in `/usr`
- ▶ Update: distributions are now making `/bin` link to `/usr/bin`, `/lib` to `/usr/lib` and `/sbin` to `/usr/sbin`. Details on https://systemd.io/THE_CASE_FOR_THE_USR_MERGE/.